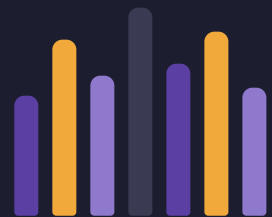




TESSERA POS · WINES & SPIRITS

Demo Walkthrough & Test Report

How to prepare and run a client demo of the whole system, role by role, with the results of the end-to-end walkthrough carried out on 27 September 2026 and the issues found and fixed.



Version: main after the demo-walkthrough merge · Environment: local demo database (APP_ENV=local)
M-PESA and KRA eTIMS run on demo drivers: no real money moves and nothing is sent to KRA.

1. What this document covers

Tessera POS is now complete for the MVP and the approved Phase 2 items: catalogue, stock ledger, till and shifts, open bottles (tots), M-PESA, purchasing, eTIMS, reports, the offline till, customers, settings, sign-in security, dashboard KPIs, credit and supplier accounts, notifications, promotions, and expenses and licences. This guide:

- prepares a demo database with realistic data (section 2);
- lists the demo logins and till PINs (section 3);
- gives a step-by-step demo script for each role (section 4);
- records what the walkthrough tested and what happened (section 5);
- lists the issues found, with their fixes (section 6), and the known limits (section 7).

Integrations are in demo mode. Real Safaricom Daraja and KRA eTIMS will be connected once the client provides credentials and the VSCU/OSCU specification. Until then, the demo M-PESA driver confirms payments with a made-up receipt number, and the demo eTIMS driver signs invoices with made-up control codes. Every screen behaves as it will in production.

2. Preparing the demo

2.1 Start the system

```
# API (port 8010) and scheduler, from backend/  
php artisan serve --host=127.0.0.1  
php artisan schedule:work  
  
# Web app (port 3010), from frontend/  
pnpm dev
```

Open **http://localhost:3010**. Use `localhost`, not `127.0.0.1`, so the sign-in cookie is shared with the API. The scheduler sends the eTIMS outbox, alerts and the daily summary. Keep it running during a demo.

2.2 Load the demo showcase

```
php artisan db:seed --class=DemoShowcaseSeeder --force
```

This works only with `APP_ENV=local` and is safe to run again before every demo. It does the following:

Item	What the seeder does
Two-step login	Switched off for the owner so the demo needs no phone. Tessera Support always keeps it. Turn it back on under Settings → Staff and roles.
Till PINs	Resets the demo PINs to those in section 3, in case they were changed while testing.
Customer credit	Switches customer credit on (any cashier may sell on account within the limit). <i>Demo Lounge & Grill</i> gets a KES 30,000 limit and 30-day terms.
Promotions	<i>Wine weekend: buy 2, 10% off</i> (any wine, 2 or more bottles). <i>Jameson happy hour</i> (KES 300 off Jameson 750 ml, 17:00–20:00). Both are approved.
Licences	The Nairobi County liquor licence expires in 20 days: it triggers an alert and is printed on the receipt. The single business permit is valid for 9 more months.

Everything else (purchase orders, goods received, sales, payments, expenses) is done in the app during the demo, so the audience sees every document being created.

3. Demo accounts

Person	Role	Back-office sign-in	Till PIN
Business Owner	Owner	owner@tessera.test / password	1470
Wanjiru Mwangi	Branch Manager	manager@tessera.test / password	4826
Otieno Kamau	Cashier	till only	2580
Amina Hassan	Cashier	till only	3691
Njeri Wambui	Storekeeper	njeri@tessera.test / password	—
Faith Achieng	Accountant	accountant@tessera.test / password	—
Tessera Support	Tessera Admin	support@tessera.test / password (two-step login)	—

These accounts exist only in local and testing databases. They are never created in production.

What each role sees

Role	Menu
Owner	Everything, including Branches, Users & roles, Settings, Messages sent and Audit log.
Branch Manager	Dashboard, POS till, sales and shifts, catalogue including promotions, all inventory, purchasing, supplier and customer accounts, M-PESA reconciliation, expenses, eTIMS, licences, reports, settings (branch level), audit log. No user administration.
Accountant	Dashboard, sales, products (view), stock views, purchasing, supplier and customer accounts, M-PESA reconciliation, expenses, eTIMS, licences, reports, audit log. No till, promotions, settings or users.
Storekeeper	Products, stock on hand, adjustments, transfers, counts, open bottles, stock ledger, purchase orders, supplier invoices and returns, suppliers. The dashboard shows stock tiles only.
Cashier	The till only (PIN sign-in on a paired till device).

4. Demo script

The script below takes about 30–40 minutes. It follows a bottle from the supplier to the customer and then to the books. Each step names who is signed in.

4.1 Owner: the business at a glance (5 min)

1. Sign in as the owner. The dashboard greets them and shows today's KPI tiles: sales, cash, M-PESA/card, gross profit, items on sale, low stock, stock value at cost, shrinkage, amounts owed by customers and to suppliers, and eTIMS invoices not yet signed.
2. Point out **Needs attention** (low stock, the licence expiring in 20 days, exceptions) and **Shifts open now**.
3. Open the bell: the liquor-licence and low-stock alerts are there.
4. Settings: show the business, receipt, till and staff sections, and how a change can be undone from its history.

4.2 Purchasing: order, receive, invoice, pay (8 min)

1. **Storekeeper or owner:** raise a purchase order for the low-stock item from the suggested supplier.

2. **Owner:** try to approve it. It is refused with "Someone else must approve this — you raised it." This is the maker-checker rule.
3. **Manager:** approve the order.
4. **Storekeeper:** receive it, with received and damaged quantities per line. A goods-received note (e.g. `MAIN-GRN-000001`) is posted and stock rises in the stock ledger.
5. **Accountant:** record the supplier invoice against the delivery. Tessera matches it to the order and the receipt.
6. **Accountant or manager:** record a supplier payment. Supplier accounts show the new balance and the payables aging.

4.3 The till: a cashier's shift (12 min)

1. Open <http://localhost:3010/till> on the paired till. Choose *Otieno* and enter PIN **2580**, then start the shift with the opening float.
2. The first screen shows best sellers. Add two bottles of wine: the *Wine weekend* promotion takes 10% off automatically. For example, $2 \times \text{KES } 1,050 - \text{KES } 210 = \text{KES } 1,890$.
3. Spirits ask the cashier to confirm the customer's age.
4. Pay by **M-PESA**: the demo prompt is confirmed with a receipt number. The receipt shows the promotion, the liquor-licence line and the (demo) eTIMS signature and QR details.
5. Choose *Demo Lounge & Grill* as the customer and pay **On account**. The sale goes on their credit account within the KES 30,000 limit.
6. **Payout:** pay KES 500 from the drawer for a small expense. A manager witnesses it with their PIN.
7. **Lock** the till. A wrong PIN is refused, the cart is kept, and Otieno's PIN unlocks it. The till also locks itself after 2 minutes idle.
8. **Return:** refund one item from an earlier sale with a reason. A return note (e.g. `MAIN-RET-000002`) is created and the stock goes back.
9. **Close the shift:** count the drawer. Tessera shows the expected cash (float + cash sales – refunds – payouts) against the counted amount and records any variance for the manager's sign-off.

4.4 Accounts and expenses (6 min)

1. **Accountant:** in *Customer accounts*, open Demo Lounge & Grill, record a payment and show the statement and aging (payments settle the oldest sales first).
2. **Accountant:** record an expense (e.g. electricity) from petty cash or the bank. It waits for approval.
3. **Owner:** an owner cannot approve an expense they raised themselves. The **manager** approves it and it appears in the expenses report.
4. **Licences & permits:** show the expiry dates, the 20-day warning and the licence line on receipts.

4.5 Reports and control (5 min)

1. Reports: sales summary, cash-ups, expenses, receivables and payables aging, stock valuation and shrinkage. Each can be printed or exported to CSV.
2. eTIMS monitor: every sale and return with its signing status. A rejected document raises an alert.
3. Audit log: every approval, price change, payout, lock and setting change, with who did it and when.
4. Messages sent: the daily summary and alerts that would go by SMS or WhatsApp (logged in demo mode).

5. Walkthrough results (27 September 2026)

The full script above was run on the local demo database through the real web app and API, signed in as each role. Results:

Area	What was done	Result
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Sign-in	Owner signs in without a code after the showcase seeder. Every role's menu matches its permissions. Requests outside the role are refused with HTTP 403.	PASS
Maker-checker	The owner could not approve their own purchase order or expense; the refusal now gives the reason. The manager approved both.	PASS
Purchasing	Order approved → received as MAIN-GRN-000001 → supplier invoice RVSD-88412 matched → supplier payment SP-000001 . The supplier balance fell to KES 92,819.20.	PASS
Till: selling	Best sellers shown. Wine promotion $2 \times 1,050 - 210 = 1,890$. Age check shown. Demo M-PESA confirmed (receipt FKF6CY3DC7). The receipt shows the promotion, the licence line and the demo eTIMS signature.	PASS
Till: on account	Sale MAIN-S-000010 of KES 3,100 went on Demo Lounge & Grill's account within the limit.	PASS
Till: payout, lock	KES 500 payout witnessed by the manager. After locking, the API refused requests (HTTP 423) and a wrong PIN was refused. After unlocking, nothing leaked into product search.	PASS
Till: return, close	Return MAIN-RET-000002 refunded KES 945. At shift close, expected cash KES 3,555 equalled counted cash, with payouts of KES 500.	PASS
Customer account	Payment MAIN-RCP-000001 of KES 1,000. Balance KES 2,100, available KES 27,900. Statement: sale 3,100, payment -1,000. Aging correct.	PASS
Expenses	MAIN-EXP-000001 approved by the manager. MAIN-EXP-000002 left pending to show the queue.	PASS
Dashboard	Owner, accountant and storekeeper dashboards load with the tiles for their role. Receivables, payables, expenses, licences and exceptions agree with the screens behind them.	PASS
Notifications	The bell showed 2 unread alerts: licence expiry and low stock.	PASS
Reports	Sales summary, cash-ups, expenses, receivables aging and payables aging all loaded. Promotions list showed KES 210 discount given.	PASS
Automated checks	After the fixes: backend suite (188 tests, 1,517 assertions, plus new checks for refusal reasons and malformed ids), Pint code style, ESLint and the UI-conventions check, TypeScript, and the production build.	PASS

6. Findings and fixes

#	Finding	Fix	Status
1	Refusals always read "You are not allowed to perform this action.", even when there was a specific reason such as "Someone else must approve this — you raised it."	The API error handler now keeps specific reasons and replaces only the framework's generic message.	FIXED
2	A malformed record id in a URL (e.g. <code>/suppliers/undefined/account</code>) gave a server error.	It now returns "Resource not found." (404).	FIXED
3	Dashboard "Shifts open now" showed only a time ("since 5:35 am") for a shift opened the day before.	The date is shown when the shift did not open today: "since 26 Sep, 5:35 am".	FIXED
4	After a sale fully on account, the till said "Paid in full".	It now says the sale went on the customer's account.	FIXED

5	The till reloaded the favourites (best sellers) several times when the screen opened.	They now load once; the offline copy is used only when the till is offline.	FIXED
6	There was no demo accountant login.	Faith Achieng (accountant@tessera.test) was added to the demo accounts.	FIXED
7	Demo till PINs had drifted during earlier testing.	The showcase seeder resets them before each demo.	FIXED
8	The demo owner is named "Business Owner", so the greeting reads "Good morning, Business".	Demo data only. Rename the owner under Users & roles before a client demo if wanted.	NOTE
9	Today's "Cash" tile can be negative when a refund of an M-PESA sale is paid in cash.	By design: refunds are paid from the drawer, and the tile shows the drawer's net cash. M-PESA refunds (B2C) are a Phase 2 item.	NOTE
10	An older shift from earlier testing (Amina, Till 1) is still open in the demo database.	Close it from the till before a client demo, or leave it to show the "open since yesterday" warning.	NOTE

7. Known limits and next steps

Waiting on the client

- Real Safaricom Daraja (STK push, C2B confirmations): needs the shortcode, passkey and API keys.
- Real KRA eTIMS (VSCU/OSCU): needs the device registration, specification and sandbox access.
- SMS/WhatsApp provider for alerts (currently logged in demo mode).

Needs a decision first

- Loyalty points: consent rules and alcohol-promotion rules need a legal check.
- Other Phase 2 / future items: M-PESA B2C refunds, debit notes, card terminals, expiry/batch tracking, excise stamps.
- Licensed hours per county: the lock is built, but each branch's hours must come from its licence.

Added after the walkthrough: automated till tests in a real browser (`pnpm e2e` : set-up, PIN, cash, promotion with M-PESA, on account, lock, licensed hours, cash-up) and the **licensed-hours lock** (Settings → Sales screen). Outside the licence's hours the till refuses alcohol; 0% ABV items and returns are unaffected.

Before going live: switch two-step login back on for owners, remove the demo accounts (production seeding never creates them), set `MPESA_DRIVER` and `ETIMS_DRIVER` to the real drivers, and run the scheduler as a service.